

Case Study: BP Deepwater Horizon, Part 2

This second case assumes that students already know the basic Macondo background. BP remained the operator of the well, but the analytical focus now shifts away from BP's internal formal organization. The question here is different. How did BP's dependence on other organizations, on external expertise, on shared and unshared data systems, on suppliers, on equipment, and on the wider regulatory environment shape the course of events?

In other words, this is a case about organization environment interaction. Macondo can be read as a temporary (inter)organizational system that had to combine contributions from multiple firms while facing uncertainty from geology, materials, diagnostics, and regulation. BP remained central, but centrality did not eliminate dependence.

A short guide to drilling and coordination terms used in this case

Only a few technical terms are needed. The main point is how organizations interacted around them.

Term	Plain meaning for this case
Mudlogging	Monitoring drilling returns and related indicators to help infer what is happening in the well.
Real time data system	A technical system that lets some actors see current operational data as work is unfolding.
Centralizers	Devices attached to casing so that the casing stays more centered in the well.
Compressive-strength analysis	Testing or analysis used to estimate when cement becomes strong enough.
Diagnostic	An extra source of information used to reduce uncertainty before proceeding.
Boundary spanning	Work that connects information or action across organizations or across organizational units.
Uncertainty absorption	The organizational process of reducing, containing, or working around what is not yet fully known.

The operation depended on a network, not on BP alone

The Macondo operation relied on multiple organizations with different contributions. Transocean supplied the rig, the drilling crew, and core offshore execution capacity. Halliburton provided cementing services, mudlogging, and technical modeling inputs. Schlumberger was available to run a cement bond log, which could have generated additional diagnostic information. Weatherford entered the picture through the effort to secure additional centralizers. Partners such as Anadarko and MOEX were part of the wider venture, but their role in the final collapse is less clear.

The important point is that the operation distributed different kinds of capability across organizational boundaries. Some firms provided equipment. Some provided data. Some provided models or diagnostic options. Some provided operational labor. The project therefore had to function as a system of

interdependence. BP could not manage the well effectively unless these external contributions were timely, visible, and interpretable.

Case Study Part 1 treated the same setting mainly as a problem of BP's formal organization. This second case asks a different question: why did a network designed to distribute competence fail to absorb uncertainty? That question directs attention to interfaces, not only to internal authority.

Exhibit 1. Major organizations and the dependencies they created

Organization	Contribution to the operation	Dependency created for BP
Transocean	Rig, drilling crew, and execution of drilling operations	BP depended on Transocean's offshore operating capacity and on its participation in interpreting conditions on the rig.
Halliburton	Cementing services, mudlogging, and technical modeling inputs	BP depended on Halliburton for specialized expertise, tests, and real time information.
Schlumberger	Potential diagnostic logging through a cement bond log	BP could reduce uncertainty only if it used the available external diagnostic service.
Weatherford	Supply of additional centralizers	BP depended on outside supply and delivery arrangements for materials relevant to the casing configuration.
Regulators and oversight bodies	Rules, approvals, and oversight context	BP operated within an institutional environment that shaped what was expected, reviewed, or left to the operator-contractor network.

Information did not move through one shared system

One of the clearest pieces of evidence concerns information architecture. Halliburton mudloggers provided real time data through an InSite system. That information could be seen by the Transocean crew, BP Well Site Leaders on the rig, and BP engineers onshore in Houston. Transocean also used its own HiTech system. BP Well Site Leaders could see HiTech as well, but Halliburton mudloggers could not, and the system was not transmitted onshore.

This matters because interorganizational coordination depends not only on formal roles but also on what different actors can actually see. Macondo operate through a layered and asymmetrical information space. That increased the need for translation and communication across firms and across locations. Under stable conditions, such an arrangement may be manageable. Under ambiguous conditions, it creates the risk that a relevant pattern visible from one position remains weakly integrated into the broader judgment process.

Public investigations points to a second information problem as well. Halliburton had conducted pre job tests and stability work on the foamed cement design, but that final compressive-strength analysis for the production casing job was still incomplete on April 19, 2010, when BP proceeded. Investigators found no evidence that cement stability results or OptiCem reports showing gas flow potential had been shared with Transocean personnel. No evidence of sharing is not identical to proof that no communication occurred in any form. But it is strong evidence that the interorganizational record was incomplete at a point when such information could have been relevant.

Exhibit 2. Selected information channels and boundary problems

Channel or tool	Who controlled or produced it	Boundary issue raised
InSite real time data	Halliburton mudlogging	Useful across firms, but still dependent on common interpretation and escalation.
HiTech system	Transocean	Created asymmetry because not all organizations, and not onshore BP, could see it.
Cement stability and OptiCem results	Halliburton technical analysis	No evidence that results were shared with Transocean personnel.
Onshore engineering input	BP engineers in Houston	Could add expertise, but only if offshore and cross-firm interpretation triggered stronger escalation.
Cement bond log option	Schlumberger diagnostic service	Extra information existed as an external option, but only if another organization chose to use it and accept delay.

Material and supply dependencies shaped the decision context

The centralizer episode shows that organization environment interaction was not only informational. It was also material and logistical. In mid April 2010, BP's team was dealing with the availability of centralizers for the casing plan. Modeling suggested that a larger number would reduce channeling risk. Yet only six centralizers with integrated stop collars were available on the rig. Additional centralizers were sought through Weatherford and moved toward a shore base in Houma. At the same time, the public record points to missing stop collars and epoxy, uncertain delivery conditions, and unresolved questions about whether all required components would arrive in usable form.

This episode matters because no single organization fully controlled the whole chain. One organization provided modeling, another provided materials, another operated the rig and schedule context, and BP had to make a choice in the middle of these dependencies. In this case, the environment was not a distant background. It was made up of supplier relationships, delivery paths, material readiness, and time coupling across firms.

Students sometimes speak about uncertainty in broad terms. Macondo shows something more concrete. Materials may exist, but not in the right place. Expertise may exist, but not in the same system. Diagnostics may be available, but only if someone accepts the operational delay needed to use them. Each dependency can look manageable on its own. Together, they can leave an organization with formal responsibility but degraded options for reducing uncertainty.

Shared interpretation across organizations remained fragile

The negative pressure test still appears in this second case, but it plays a different analytical role than in Case Study 1. Here, the test matters less as proof of BP's formal authority and more as an example of cross-organizational interpretation under uncertainty. The court record indicates that anomalies were present, yet the test was treated as successful and work continued. BP held final authority, but Transocean personnel were involved in the discussion, and BP engineers onshore were part of the wider interpretive setting.

Organizations often try to absorb uncertainty by dividing labor, assigning expertise, and using tests as decision gates. Such arrangements work only if the outputs of those tests travel credibly across boundaries and if anomalies trigger renewed inquiry rather than a shared move toward continuation. Macondo suggests how distributed interpretation can stabilize around a problematic reading when data, expertise, and authority are spread unevenly across a temporary network.

The dossier also points to additional limits in uncertainty absorption. Some limits were diagnostic, such as the decision not to use the cement bond log. Some were informational, such as the incomplete compressive-strength analysis and the incomplete documented sharing of some cement-related results. Some were material, such as the centralizer supply problem. Some were technical, such as later-documented reliability problems in the blowout preventer system. Each sat at a different interface, and each involved a different set of organizations.

Exhibit 3. Sources of uncertainty and incomplete responses

Source of uncertainty	Attempt to reduce it	Why the attempt remained limited
Changing geological conditions	Ongoing plan adaptation and formal change management	Investigators later criticized the adequacy of risk evaluation for several major changes.
Unfinished cement analysis	Technical review before proceeding	BP proceeded while final compressive-strength analysis was still incomplete.
Potential gas flow shown in modeling	Cross-organizational communication of technical results	BOEMRE found no evidence that certain results were shared with Transocean personnel.
Need for more information about cement placement	Use of the cement bond log	The diagnostic service was available but not used.
Emergency equipment reliability	Dependence on the blowout preventer system	Later investigations documented technical reliability problems that the network had not fully absorbed.

The wider regulatory environment also mattered

Macondo also invites a broader view of organization environment interaction. The project did not operate only in relation to contractors and suppliers. It also operated within an institutional environment shaped by offshore regulation. Governmental bodies created a rule after the disaster, which formalized stronger expectations around safety and environmental management, documentation, contractor selection, and operator-contractor alignment on safety practices.

Students should not read later rules backward as if those exact requirements had already governed the operation on April 20, 2010. The more careful lesson is different. The disaster made visible how much coordination and uncertainty management had been left to the operator-contractor network itself. In that sense, the institutional environment matters because it helps explain why so much depended on the network's own ability to manage coordination.

Closing

Seen through this second lens, the Macondo blowout was not only the failure of BP's internal formal control. It was the breakdown of a interorganizational system that had to coordinate authority, expertise, diagnostics, materials, equipment, and oversight under uncertainty. BP was central to that system, but it did not operate alone and it did not face an environment it could fully control.

For students of organization, the central task is to explain why a network designed to distribute competence and absorb uncertainty instead left critical uncertainties unresolved. Which dependencies were manageable, which became dangerous, and why were possible corrective mechanisms not activated in time?

Orienting Questions for Students

1. Which interorganizational dependencies were most consequential at Macondo?
2. Where does the case show attempts to absorb or contain uncertainty across organizational boundaries, and why do those attempts appear ineffective?
3. What can we learn from this case for other organizations operating under uncertain conditions?

Sources Used for This Case

Bureau of Ocean Energy Management, Regulation and Enforcement. (2011, September 14). Report regarding the causes of the April 20, 2010 Macondo well blowout. U.S. Department of the Interior.

Deepwater Horizon Study Group. (2011, March). Final report on the investigation of the Macondo well blowout. Center for Catastrophic Risk Management, University of California, Berkeley.

Federal Register. (2010, October 15). Safety and environmental management systems (SEMS) final rule, 30 CFR Part 250.

National Academy of Engineering & National Research Council. (2012). Macondo well, Deepwater Horizon blowout: Lessons for improving offshore drilling safety. National Academies Press.

U.S. Chemical Safety and Hazard Investigation Board. (2014). Investigation report, volume 1: Explosion and fire at the Macondo well (Report No. 2010-10-I-OS).

U.S. Chemical Safety and Hazard Investigation Board. (2014). Investigation report, volume 2: Drilling rig explosion and fire at the Macondo well, BOP technical failure analysis.

U.S. Chemical Safety and Hazard Investigation Board. (2016). Investigation report, volume 4: Regulatory oversight of U.S. offshore oil and gas operations.

United States District Court, Eastern District of Louisiana. (2014, September 9). In re Oil Spill by the Oil Rig Deepwater Horizon on April 20, 2010, revised findings of fact and conclusions of law, phase one.

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